

02.5 Evidence Quality and Readiness

Coding Protocol: Evidence Quality and Implementation Readiness Rubrics

This protocol documents the two evaluative ratings applied to every extraction record: the Evidence_Quality tier rating and the Implementation_Readiness classification. The two are applied independently and for different purposes. Evidence quality records the methodological trustworthiness of the finding. Implementation readiness records the degree to which the tool has been tested or deployed in real gambling environments, regardless of how well that testing was conducted. A tool may be deployed at scale without being well studied; a tool may be studied rigorously without being deployed.

1. Evidence Quality Rubric

Evidence quality is rated on a five-level scale (HIGH, MODERATE-HIGH, MODERATE, LOW-MODERATE, LOW) anchored to methodological trustworthiness: the degree to which design features warrant confidence in the findings. The anchoring features are randomisation, objective behavioural outcomes, sample size, follow-up duration, pre-registration, multi-site replication, measurement validity, and attrition.

The rating is applied to the study's primary finding as extracted, not to the paper as a whole, and not to the tool itself. A single paper evaluating multiple tools may produce records with different quality ratings if the evidentiary basis for each tool differs (for example, one tool tested in a randomised trial and another described only on a programme-audit basis). The rating is paired with a direction indicator (see 02.4 Evidence Direction and Sample Size) which captures the valence of the finding independently of its quality.

1.1 Tier Definitions

HIGH ($n = 0$; not attested in this review)

Definition. The highest-quality evidence tier, reserved for findings backed by randomised design with objective behavioural outcomes, extended follow-up, multi-site replication, validated measures, low attrition, and pre-registration.

Inclusion. Randomised design (RCT or field experiment with random allocation); objective behavioural outcomes extracted from operator account data or validated physiological/behavioural measures; follow-up of at least six months; evidence from multiple independent operators or jurisdictions; use of validated psychometric instruments where self-report measures are included; attrition rates within conventional benchmarks; pre-registration of hypotheses and analytic plan.

Exclusion. Studies meeting most but not all criteria ($\rightarrow$ MODERATE-HIGH). No records in the current review met all HIGH criteria; this reflects the state of the field rather than a feature of the rubric.

Example. Not attested in this review. The HIGH tier is retained in the rubric to preserve the ceiling of the scale and to make explicit what the strongest achievable evidence in the field would look like.

MODERATE-HIGH ($n = 14$; 3.4% of records)

Definition. Strong evidence meeting most but not all HIGH criteria. Typically an RCT or well-controlled field experiment with objective outcomes and meaningful follow-up, but with at least one substantive limitation: single operator or jurisdiction, absence of pre-registration, shorter follow-up, or partial attrition management.

Inclusion. RCTs with matched or randomised controls and objective behavioural outcomes; prospective longitudinal designs with independent replication samples; well-powered randomised

field experiments; studies integrating objective behavioural data with validated psychometric measures.

Exclusion. Studies with an additional major methodological limitation such as self-selection into the tool, lab-only data, or absence of a control condition (→ MODERATE). Studies meeting all HIGH criteria (→ HIGH).

Example. MODERATE-HIGH (positive): RCT with matched controls showing 30% reduction in theoretical loss over 12 months in the intervention group versus 7% in controls; objective behavioural outcomes; follow-up at multiple time points (Jonsson et al., 2021).

MODERATE (*n* = 201; 49.0% of records)

Definition. Sound methodology with meaningful but manageable limitations. Typically a single-operator field study with objective behavioural data and a defensible comparison condition, but with limitations in generalisability, self-selection, or causal identification.

Inclusion. Large-scale behavioural tracking with objective transaction data but from a single operator; pre-post designs with propensity score matching or other non-randomised causal controls; well-designed laboratory experiments with external validity concerns; observational studies with adequate sample and clear analytic strategy but without randomised controls.

Exclusion. Studies with substantial limitations such as small student samples, lab-only settings with brief exposure, or small effect sizes without replication (→ LOW-MODERATE). Studies meeting the RCT threshold and objective-outcome criteria (→ MODERATE-HIGH).

Example. MODERATE (positive): Pre-post evaluation of mandatory play breaks using large-scale operator transaction data with propensity matching; objective behavioural outcomes; 12-month follow-up; limitations include self-selection and single operator.

LOW-MODERATE (*n* = 185; 45.1% of records)

Definition. Acceptable methodology with substantial limitations that constrain confidence in the finding. Typically a lab-based experimental study with a student sample and brief exposure, or a field study with small effect sizes and limited follow-up.

Inclusion. Randomised laboratory experiments with student samples and brief exposure to simulated gambling; field deployments with objective outcomes but small effect sizes, short follow-up, or single-operator single-venue constraints; cross-sectional studies with large samples but limited causal inference; qualitative studies with small samples but systematic analysis.

Exclusion. Studies with no control group, no comparison condition, or proof-of-concept designs that cannot rule out non-specific factors (→ LOW). Studies with a single substantive limitation but otherwise strong methodology (→ MODERATE).

Example. LOW-MODERATE (positive): Pop-up messages reduced session expenditure in a laboratory simulation; randomised controlled design with objective behavioural outcome; limitations include lab setting with student sample, brief exposure, and single-session simulation rather than real-money gambling.

LOW (*n* = 10; 2.4% of records)

Definition. Substantial methodological limitations that limit causal inference. Typically a pre-post study without a control group, a proof-of-concept evaluation with small sample, or a feasibility study where the primary contribution is demonstrating implementability rather than effect.

Inclusion. Pre-post designs without control condition; proof-of-concept or pilot studies with small samples; feasibility studies reporting uptake and acceptability without outcome comparison; studies where natural recovery cannot be distinguished from tool effects.

Exclusion. Studies with a defensible comparison condition or propensity-matched control (→ LOW-MODERATE).

Example. LOW (mixed): Small laboratory study with student sample showing that personalised feedback increased message attention in eye-tracking but did not produce differences in gambling behaviour; pre-post design without control for selection effects.

2. Implementation Readiness Rubric

Implementation readiness records where along the research-to-practice pipeline a tool currently sits. Four levels are used: Deployed, Validated, Tested, and Conceptual. The levels are defined by the setting in which the tool has been empirically evaluated, not by the rigour of that evaluation. Quality is recorded separately in the Evidence_Quality field.

A tool may be Deployed under regulatory mandate without strong effectiveness evidence; a tool may be rigorously Validated in a field experiment without subsequent scaled deployment. Both facts are informative and are captured independently. The combination of readiness and quality determines the translational status: Deployed + MODERATE-HIGH evidence is implementation-ready and well-supported; Tested + MODERATE is experimentally supported but unimplemented; Deployed + LOW-MODERATE is widespread-but-weakly-supported practice.

2.1 Readiness Level Definitions

Deployed (*n* = 148; 36.1% of records)

Definition. The tool is operating in a real gambling environment as a standard feature, regulatory requirement, or active operator practice. Evidence comes from the deployment context itself, whether by regulatory mandate, operator implementation, or policy audit documenting the tool in active use.

Inclusion. Tools in active use by named operators (Norsk Tipping, Svenska Spel, Holland Casino); mandatory regulatory requirements in force at the time of the study; tools audited as operational in a licensed jurisdiction; cross-operator deployment documented by regulatory review.

Exclusion. Tools evaluated in real gambling settings but not operationally deployed (→ Validated); tools tested in laboratory settings (→ Tested); tools described or theorised without empirical evaluation (→ Conceptual).

Example. Deployed (mandatory regulatory requirement): Deposit limit tool available to all account holders at a licensed online operator under regulatory requirement; operator transaction data document routine use.

Validated (*n* = 85; 20.7% of records)

Definition. The tool has been empirically evaluated in a real gambling environment (real venues, real money, real players) but is not necessarily in routine operational use. This level captures field-experimental evaluation and real-world trials.

Inclusion. Randomised field experiments in licensed gambling venues; real-money RCTs conducted with operator collaboration; pragmatic trials embedded in live gambling products; naturalistic trials with real players, real money, and real gambling consequences.

Exclusion. Evaluations in simulated or laboratory environments regardless of design rigour (→ Tested); tools in routine operational deployment (→ Deployed).

Example. Validated (field-tested in real venues): Telephone-delivered motivational interviewing intervention for high-spending players, randomised field experiment conducted with an online operator; real money, real players, 12-month follow-up.

Tested (*n* = 87; 21.2% of records)

Definition. The tool has been evaluated in a laboratory or simulated gambling environment but has not been tested with real money or real gambling consequences. This level captures well-designed experimental work that establishes proof of mechanism without establishing proof of real-world effectiveness.

Inclusion. Laboratory studies with simulated EGMs, VLTs, or card games; student samples playing for points or credits rather than money; simulated online gambling platforms; experimental studies with operant analogues of gambling behaviour.

Exclusion. Studies using real money or real gambling products (→ Validated); studies describing a tool without empirical test (→ Conceptual); tools in operational use regardless of test setting (→ Deployed).

Example. Tested (lab, simulated VLT): Randomised experiment using simulated VLT task with student participants receiving credits; pop-up message manipulation during play; behavioural outcomes measured within the laboratory session.

Conceptual (*n* = 90; 22.0% of records)

Definition. The tool is described, theorised, or derived from other contexts, but has no direct empirical evaluation in a gambling setting. This level captures design principles, policy proposals, and transfer claims from adjacent fields.

Inclusion. Design principles inferred from laboratory cognitive tasks without gambling application; policy proposals without deployment or trial; theoretical transfer of tools validated in alcohol or tobacco contexts; qualitative or consultative work proposing tools without evaluating them; low-risk gambling guidelines derived from epidemiological data but not implemented as deployed tools.

Exclusion. Any tool with direct empirical evaluation in a gambling setting, at any level of rigour (→ Tested, Validated, or Deployed as appropriate).

Example. Conceptual (design principle from lab study): Transfer of stop-signal response inhibition training paradigm from alcohol contexts to gambling, proposed as a cognitive-training tool but not yet evaluated in a gambling population.

3. Independence of the Two Ratings

The rubrics are applied independently. An extraction's evidence quality rating is determined entirely by the methodological features of the study; its implementation readiness is determined entirely by the environment in which the tool operates or was tested. The full distribution of records across the two ratings is given in Table 02.5.1.

Table 02.5.1

Evidence Quality by Implementation Readiness

Quality level	Deployed	Validated	Tested	Conceptual	Total
MODERATE-HIGH	4	2	4	4	14
MODERATE	77	40	39	45	201
LOW-MODERATE	60	42	43	40	185
LOW	7	1	1	1	10
Total	148	85	87	90	410

Note. Cells show record counts. The HIGH quality tier is omitted (no records attested). Column percentages reflect implementation distributions; row percentages reflect quality distributions. The Deployed × MODERATE-HIGH cell (*n* = 4) is the smallest of the high-readiness/high-quality combinations and represents 1.0% of the 410-record evidence base.

Of the 148 Deployed records, 4 are rated MODERATE-HIGH and 77 are rated MODERATE; the remainder are LOW-MODERATE (60) or LOW (7). This pattern indicates that widely-used tools are not uniformly supported by the strongest evidence: most deployed tools sit in the MODERATE tier rather than the MODERATE-HIGH tier.

Of the 14 MODERATE-HIGH records, 4 are Deployed, 2 are Validated, 4 are Tested, and 4 are Conceptual. The strongest-quality evidence in the review is distributed across all four readiness levels rather than concentrated at the Deployed or Validated end. The Deployed × MODERATE-HIGH combination accounts for only 4 of 410 records (1.0%), reflecting a structural gap between what has been rigorously evaluated and what is in routine operational use.

3.1 Using the Ratings Together

For decision-supporting analyses (for example, the convergent evidence set in [OSF: 03_2_Convergent_Evidence_Set.xlsx]), both ratings are used as filter criteria. A tool is considered decision-supporting when it is Validated or Deployed (established in real gambling environments) and is rated MODERATE or MODERATE-HIGH on evidence quality. This joint filter isolates the subset of the evidence base where both implementation status and methodological confidence are sufficient to inform practice recommendations.

For scoping analyses (for example, the overviews in Tables 2 and 3 of the main manuscript), both ratings are reported but neither is used as an exclusion criterion. The scoping purpose is to document the full landscape of tools that have been proposed, evaluated, or deployed in the literature, including tools with weaker evidence and tools that remain conceptual. The decision-supporting filter is applied only where actionable claims are drawn from the review.